

TransPNPC: Transformer-Based Deep Unfolding for Phase Noise Parameter Compensation in Beamspace DoA Estimation

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Abstract—In this paper, we propose TransPNPC, a transformer-based deep unfolding framework that mimics the iterative phase noise parameter compensation (PNPC) procedure to recover a phase-error-free signal subspace for beamspace direction-of-arrival (DoA) estimation. Each unfolding stage corresponds to one iteration of a PNPC-DFT-ESPRIT algorithm, where a transformer learns the phase noise parameter update and compensates the distorted signal subspace. The unfolded architecture is fully differentiable and can be combined with subspace-based DoA estimators such as DFT-ESPRIT. Our simulation results demonstrate that TransPNPC outperforms the conventional iterative PNPC-DFT-ESPRIT algorithm in the low SNR region, and has a smaller computational complexity than the iterative PNPC-DFT-ESPRIT algorithm.

Index Terms—Phase noise compensation, model-based deep unfolding, beamspace ESPRIT, DoA estimation.

I. INTRODUCTION

Hybrid beamforming architectures provide a trade-off between beamforming performance, hardware cost, and power consumption. Therefore, they have been widely adopted in millimeter-wave (mmWave) and massive MIMO systems. In such systems, beamspace direction-of-arrival (DoA) estimation enables high-resolution sensing with a limited number of RF chains by projecting the antenna space measurements onto a low-dimensional beamspace representation. To increase the number of coherent beamspace measurements, time-domain-multiplexed (TDM) beam switching is often employed, which potentially introduces random phase jump errors caused by hardware imperfections [1]. These phase errors destroy the signal subspace and significantly degrade the performance of classical subspace-based DoA estimation methods, such as multiple signal classification (MUSIC) or estimation of signal parameters via rotational invariant techniques (ESPRIT).

To mitigate the impact of phase errors, compressive sensing (CS)-based methods have been investigated by exploiting the sparsity of mmWave channels [2]–[4]. However, CS-based approaches suffer from grid mismatch, a high computational complexity, and calibration problems in practical systems. More recently, the phase noise parameter compensation (PNPC) DFT-ESPRIT algorithm [5] has been proposed to

jointly estimate the DoAs and the phase errors by extending the beamspace shift-invariance equation and performing an alternating iterative optimization procedure. Although PNPC can asymptotically approach the performance of ideal coherent measurements, i.e., without the impact of phase errors, its iterative null-space estimation introduces an additional computational complexity, particularly in scenarios with a single RF chain.

Meanwhile, deep learning (DL)-based DoA estimation methods have gained considerable interest due to their strong capability to learn nonlinear mappings from noisy measurements to DoAs. Nevertheless, purely data-driven approaches often lack interpretability. To address this limitation, the paradigm of model-based deep learning has been proposed, which integrates physical signal models with data-driven learning [6]. A transformer-aided noise subspace learning framework (TransMUSIC) [7] estimates the noise subspace from low-resolution quantized measurements and combines it with MUSIC for DoA estimation. However, the learned noise subspace is a data-driven latent representation rather than a physical noise subspace.

In this paper, we propose a transformer-aided phase noise parameter compensation (TransPNPC) framework, which converts the iterative PNPC-DFT-ESPRIT algorithm into a deep unfolding architecture. Instead of explicitly performing null-space estimation and iterative least-squares updates, TransPNPC employs a transformer encoder [8] to learn the phase-error estimation in a data-driven manner, while preserving the model-based phase compensation and ESPRIT processing steps. By unfolding the PNPC iterations into a neural network structure, TransPNPC achieves fast convergence and a reduced computational complexity.

The main contributions of this work can be summarized as follows. We propose a novel transformer-based unfolding architecture for joint phase-error compensation and DoA estimation in the DFT beamspace. Moreover, we integrate the transformer-based phase-error estimator with model-based PNPC compensation, and develop a hybrid learning framework. Our simulations demonstrate that the proposed TransP-

NPC achieves a better DoA estimation accuracy than PNP-C-DFT-ESPRIT in the low-SNR region with a lower computational complexity.

Notation: Non-bold letters, lower-case and upper-case bold-faced letters denote scalars, vectors, and matrices, respectively. Moreover, the symbols $(\cdot)^H$, $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^+$ denote conjugate transpose, transpose, conjugate, and pseudo-inverse, respectively. The operator $\otimes$ is the Kronecker product, and $\diamond$ is the Khatri-Rao product. The operator $\text{diag}(\mathbf{x})$ represents a diagonal matrix constructed from the elements of $\mathbf{x}$, and the operator $\|\mathbf{A}\|_F$ computes the Frobenius norm of the matrix $\mathbf{A}$. For a vector $\mathbf{a}$, $[\mathbf{a}]_m$ is its m^{th} entry, whereas for a matrix $\mathbf{A}$, $[\mathbf{A}]_{[m,n]}$ represents its $(m,n)^{\text{th}}$ entry. The operator $\angle(\cdot)$ takes the angle of a complex number, $\text{vec}(\cdot)$ is the column-wise matrix vectorization operator. Furthermore, $\mathbf{0}_m$ represents a $m \times 1$ vector with all 0 entries, $\mathbf{1}_m$ stands for a $m \times 1$ vector with all 1 entries, $\mathbf{I}_m$ denotes a $m \times m$ identity matrix, and $\mathbb{E}\{\cdot\}$ denotes the expectation operator.

II. PROBLEM FORMULATION AND ITERATIVE PNP-C-DFT-ESPRIT ALGORITHM

A. Problem Formulation

We consider a hybrid beamforming receiver equipped with a uniform linear array (ULA) of M antenna elements and L RF chains. To obtain B coherent DFT beamspace measurements, time-domain multiplexing (TDM) is performed within the channel coherence time. The system model and notation follow [5] and are briefly summarized here for completeness.

For the n -th snapshot, the received signal in the antenna element space is given by

$$\mathbf{h}_n = \mathbf{A}\mathbf{x}_n + \mathbf{z}_n \in \mathbb{C}^{M \times 1}, \quad (1)$$

where $\mathbf{A} = [\mathbf{a}(\mu_1), \dots, \mathbf{a}(\mu_d)] \in \mathbb{C}^{M \times d}$ is the array steering matrix, and $\mu_i = \frac{2\pi\Delta}{\lambda} \sin(\theta_i)$ denotes the spatial frequency for the i^{th} path, and λ is the wavelength. The array steering vector is given by $\mathbf{a}(\mu_i) = [1, e^{j\mu_i}, \dots, e^{j(M-1)\mu_i}]^T \in \mathbb{C}^{M \times 1}$, and $\mathbf{z}_n \in \mathbb{C}^{M \times 1}$ denotes zero-mean circularly symmetric complex Gaussian noise with variance σ_z^2 .

During the t -th ($t \in \{1, 2, \dots, T\}$) TDM measurement, the received signal is projected into the DFT beamspace using the subcodebook $\mathbf{W}_{B,t} \in \mathbb{C}^{M \times B}$, whose columns are selected from an $M \times B$ DFT matrix $\mathbf{W}_B$ to form a sector of interest (SoI). Such a SoI can be obtained as proposed in [9]. Due to hardware imperfections, a random phase jump error is introduced and is modeled as

$$\boldsymbol{\Omega}^{(t)} = e^{j\omega_t} \mathbf{I}_L \in \mathbb{C}^{L \times L}, \quad (2)$$

where $\omega_t \in [-\pi, \pi]$ denotes the random phase error associated with the t -th TDM measurement. Accordingly, the resulting beamspace observation for the n -th snapshot is given by

$$\tilde{\mathbf{y}}_n^{(t)} = \boldsymbol{\Omega}^{(t)} \mathbf{W}_{B,t}^H \mathbf{A} \mathbf{x}_n + \boldsymbol{\Omega}^{(t)} \mathbf{W}_{B,t}^H \mathbf{z}_n^{(t)} \in \mathbb{C}^{L \times 1}. \quad (3)$$

To cover the entire SoI that is formed by $\mathbf{W}_B$, the analog phase shifters are updated to switch between different subcodebooks. Without loss of generality, we assume that $T = \frac{B}{L}$. The

projection is achieved by TDM measurements over T OFDM symbols. Stacking all T TDM measurements yields the B -dimensional beamspace snapshot

$$\tilde{\mathbf{y}}_n = \begin{bmatrix} \tilde{\mathbf{y}}_n^{(1)} \\ \vdots \\ \tilde{\mathbf{y}}_n^{(T)} \end{bmatrix} = \boldsymbol{\Omega}_P \mathbf{W}_B^H \mathbf{A} \mathbf{x}_n + \tilde{\mathbf{z}}_n \in \mathbb{C}^{B \times 1}, \quad (4)$$

where $\mathbf{W}_B = [\mathbf{W}_{B,1}, \dots, \mathbf{W}_{B,T}] \in \mathbb{C}^{M \times B}$, $\boldsymbol{\Omega}_P = \boldsymbol{\Omega}_T \otimes \mathbf{I}_L \in \mathbb{C}^{B \times B}$, $\boldsymbol{\Omega}_T = \text{diag}(\boldsymbol{\omega}_T) \in \mathbb{C}^{T \times T}$, and $\boldsymbol{\omega}_T = [e^{j\omega_1}, \dots, e^{j\omega_T}] \in \mathbb{C}^{T \times 1}$ denotes the phase error vector.

After collecting N snapshots, the stacked beamspace measurement matrix is

$$\tilde{\mathbf{Y}} = \boldsymbol{\Omega}_P \mathbf{W}_B^H \mathbf{A} \mathbf{X} + \tilde{\mathbf{Z}} \in \mathbb{C}^{B \times N}. \quad (5)$$

The objective is to jointly estimate the phase errors and the DoAs from the measurement matrix $\tilde{\mathbf{Y}}$ in DFT beamspace.

B. Iterative PNP-C-DFT-ESPRIT Algorithm

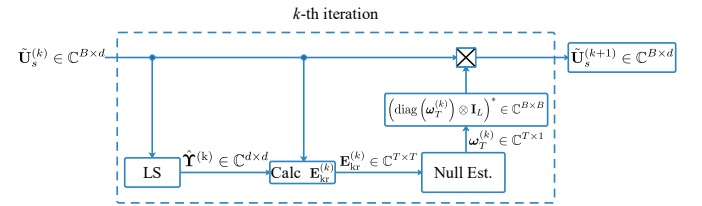


Fig. 1: PNP-C-DFT-ESPRIT algorithm diagram

The iterative PNP-C-DFT-ESPRIT algorithm jointly estimates the phase errors and the spatial frequencies via an alternating optimization procedure. The key insight is that, under the extended beamspace shift-invariance relation, the phase-error vector $\boldsymbol{\omega}_T$ lies in the null-space of a Kronecker-structured matrix constructed from the phase-compensated signal subspace [5].

Fig. 1 depicts the algorithmic flow at the k -th iteration. Given the current estimate of the beamspace signal subspace matrix $\tilde{\mathbf{U}}_s^{(k)} \in \mathbb{C}^{B \times d}$ (note that at the first iteration, the columns of $\tilde{\mathbf{U}}_s^{(1)}$ are obtained as the d dominant left singular vectors of the phase error distorted beamspace measurement matrix $\tilde{\mathbf{Y}}$), the Least Squares estimate of the matrix $\hat{\mathbf{Y}}^{(k)}$ is obtained as

$$\hat{\mathbf{Y}}^{(k)} = (\mathbf{G}_{1,B} \tilde{\mathbf{U}}_s^{(k)})^+ \mathbf{G}_{2,B} \tilde{\mathbf{U}}_s^{(k)}, \quad (6)$$

where $\mathbf{G}_{1,B}$ and $\mathbf{G}_{2,B}$ are the pre-defined selection matrices corresponding to the SoI [5]. Based on $\tilde{\mathbf{U}}_s^{(k)}$ and $\hat{\mathbf{Y}}^{(k)}$, a Kronecker-structured matrix $\mathbf{M}_{kr}^{(k)} \in \mathbb{C}^{d(B-1) \times T}$ is constructed as

$$\mathbf{M}_{kr}^{(k)} = \left([(\tilde{\mathbf{U}}_s^{(k)} \hat{\mathbf{Y}}^{(k)})^T \diamond \mathbf{G}_{1,B}] - [\tilde{\mathbf{U}}_s^{(k)T} \diamond \mathbf{G}_{2,B}] \right) (\mathbf{I}_T \otimes \mathbf{1}_L), \quad (7)$$

where the residual phase-error vector $\boldsymbol{\omega}_T^{(k)}$ resides in its null-space. Specifically, the estimated residual phase error vector $\boldsymbol{\omega}_T^{(k)} \in \mathbb{C}^{T \times 1}$ satisfies

$$\mathbf{M}_{kr}^{(k)} \boldsymbol{\omega}_T^{(k)*} = \mathbf{0}_{d(B-1)}. \quad (8)$$

Extensive numerical simulations in [5] show that the null-space of $\mathbf{M}_{\text{kr}}$ is one-dimensional. A theoretical proof of the one-dimensional null-space property will be investigated in our future work.

To reduce the computational burden of null-space estimation, we consider the Gram matrix

$$\mathbf{E}_{\text{kr}}^{(k)} = \mathbf{M}_{\text{kr}}^{(k)\text{H}} \mathbf{M}_{\text{kr}}^{(k)} \in \mathbb{C}^{T \times T}, \quad (9)$$

which shares the same null-space as $\mathbf{M}_{\text{kr}}^{(k)}$. With additive noise and a finite number of snapshots N , the null-space basis vector is estimated as the eigenvector $\mathbf{v}_{\min}$ corresponding to the smallest eigenvalue of $\mathbf{E}_{\text{kr}}^{(k)}$. Since the null-space solution has an arbitrary scaling, we normalize it by a unit-modulus operation,

$$\omega_T^{(k)} = e^{j \cdot \angle(\mathbf{v}_{\min})}, \quad (10)$$

where the operator $\angle(\cdot)$ is taken elementwise.

Finally, $\omega_T^{(k)}$ is used to compensate the beamspace signal subspace for the next iteration, which is given by

$$\tilde{\mathbf{U}}_s^{(k+1)} = \left(\text{diag}(\omega_T^{(k)}) \otimes \mathbf{I}_L \right)^* \tilde{\mathbf{U}}_s^{(k)}. \quad (11)$$

By alternately updating the signal subspace and the residual phase-error vector, the PNPC-DFT-ESPRIT algorithm jointly compensates the distorted signal subspace matrix $\tilde{\mathbf{U}}_s^{(k)}$ and converges to the phase-error-free signal subspace. Then, the classical DFT-ESPRIT algorithm is applied to estimate the DoAs.

III. PROPOSED TRANSPNPC ARCHITECTURE

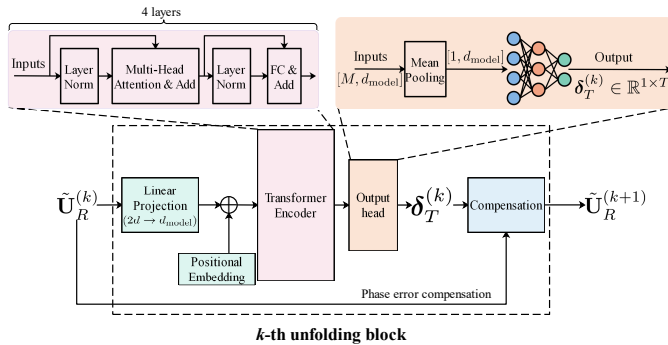


Fig. 2: k -th unfolding block of TransPNPC

In this section, we present the proposed TransPNPC architecture, which builds upon the iterative PNPC-DFT-ESPRIT algorithm and converts its model-based alternating optimization procedure into a transformer-based deep unfolding architecture, where each unfolding block corresponds to one iteration of the original PNPC algorithm. Instead of explicitly performing null-space estimation and iterative phase-error updates as in PNPC-DFT-ESPRIT, TransPNPC uses a transformer encoder to learn the phase-error estimation in a data-driven manner. The overall architecture of TransPNPC is illustrated in Fig. 2, where each unfolding block corresponds to one iteration of the original PNPC-DFT-ESPRIT algorithm.

A. Architecture

Within each unfolding block, the real and imaginary parts of the current compensated signal subspace matrix $\tilde{\mathbf{U}}_s^{(k)} \in \mathbb{C}^{B \times d}$ are concatenated as $\mathbf{U}_R^{(k)} = [\text{Re}\{\tilde{\mathbf{U}}_s^{(k)}\}, \text{Im}\{\tilde{\mathbf{U}}_s^{(k)}\}] \in \mathbb{R}^{B \times 2d}$ and interpreted as a sequence of B tokens, each with a $2d$ -dimensional feature representation. These features are mapped into a d_{model} -dimensional latent space via a linear projection, followed by a positional embedding module to preserve the structural ordering of the beamspace components. The embedded tokens are then fed into a 4-layer transformer encoder to capture both local and global dependencies along the beamspace dimension.

The outputs of transformer encoder are fed into an output head, which yields the estimated residual phase error parameter vector $\delta_T^{(k)} \in \mathbb{R}^{1 \times T}$ and the phase error vector $\omega_T^{(k)} = e^{j\delta_T^{(k)}} \in \mathbb{C}^{1 \times T}$, where TransPNPC replaces the model-based PNPC operations in (6)–(10) with a data-driven transformer module. The estimated phase-error vector is then applied in a model-based phase compensation module to update the signal subspace, as shown in Fig. 1 and (11), which yields the updated matrix $\tilde{\mathbf{U}}_R^{(k+1)}$ for the next unfolding block.

B. Training Procedure

The proposed TransPNPC is trained in an end-to-end supervised manner. The training set consists of Q pairs of phase error and additive noise distorted beamspace measurements $\tilde{\mathbf{Y}}^{(q)} \in \mathbb{C}^{B \times N}$ ($q = 1, 2, \dots, Q$) and noise-free and phase-error-free beamspace measurements $\mathbf{Y}^{(q)} \in \mathbb{C}^{B \times N}$, i.e., $\mathcal{D}_{\text{train}} = \{\tilde{\mathbf{Y}}^{(q)}, \mathbf{Y}^{(q)}\}_{q=1}^Q$. The columns of $\tilde{\mathbf{U}}_s^{(k)}$ as input to the TransPNPC and the columns of the matrix $\mathbf{U}_s$ used for the loss computation are obtained as the d dominant left singular vectors of $\tilde{\mathbf{Y}}^{(q)}$ and $\mathbf{Y}^{(q)}$, respectively. TransPNPC performs K unfolding steps, where each unfolding block predicts the residual phase error parameter vector $\delta_T^{(k)}$ and updates the matrix that spans the estimated signal subspace $\tilde{\mathbf{U}}_R^{(k)}$ in a model-based phase compensation module. The training loss function is defined as

$$\text{Loss} = \|\mathbf{P}_{\text{est}} - \mathbf{P}_{\text{true}}\|_F^2, \quad (12)$$

where $\mathbf{P}_{\text{est}} = \tilde{\mathbf{U}}_s^{(K+1)} \tilde{\mathbf{U}}_s^{(K+1)\text{H}}$, $\mathbf{P}_{\text{true}} = \mathbf{U}_s \mathbf{U}_s^{\text{H}}$. K denotes the total number of unfolding blocks. The projector loss is used because it uniquely determines the subspace. Specifically, if $\mathbf{P}_{\text{est}} = \mathbf{P}_{\text{true}}$, then $\text{span}(\tilde{\mathbf{U}}_s^{(K+1)}) = \text{span}(\mathbf{U}_s)$. Hence, there exists a nonsingular matrix $\mathbf{T}_K$ such that $\tilde{\mathbf{U}}_s^{(K+1)} = \mathbf{U}_s \mathbf{T}_K$. The phase-error-free shift-invariance equation is given by $\mathbf{G}_{1,B} \mathbf{U}_s \mathbf{T}_K = \mathbf{G}_{2,B} \mathbf{U}_s$. Substituting $\tilde{\mathbf{U}}_s^{(K+1)} = \mathbf{U}_s \mathbf{T}_K$, we obtain $\mathbf{G}_{1,B} \tilde{\mathbf{U}}_s^{(K+1)} (\mathbf{T}_K^{-1} \mathbf{T}_K) = \mathbf{G}_{1,B} \tilde{\mathbf{U}}_s^{(K+1)}$. Therefore, the shift-invariance structure is preserved.

IV. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed TransPNPC algorithm through Monte-Carlo simulations. We consider a uniform linear array (ULA) with half-wavelength inter-element spacing. The training and testing setup are

TABLE I: Simulation Parameters

Parameter	Value
Antenna spacing	$\Delta = \lambda/2$
Num. sources / antennas	$d = 3, M = 16$
Num. beams / snapshots	$B = 16, N = 100$
Num. RF chains / source correlation	$L = 1, \rho = 0$
Phase error parameter	$\omega_t \sim \mathcal{U}(-\pi, \pi)$
Training DoAs (deg)	$\theta_i \sim \mathcal{U}(-60^\circ, 60^\circ)$
Training / Validation samples	200,000 / 20,000
Training SNR	0dB, 20 dB
Test DoAs (deg)	$\theta = [20^\circ, 28^\circ, 30^\circ]$
Test trials	1,000
Embedding dimension d_{model}	32
Num. attention heads	4
FFN hidden dimension	128
Batch size / Learning rate	256 / 10^{-4}
Optimizer	AdamW [10]

summarized in Table I. During training, the DoAs are randomly drawn from a uniform distribution over $[-60^\circ, 60^\circ]$, while fixed off-grid DoAs are used during testing. A hybrid beamforming architecture with a single RF chain is considered, and the sources are assumed to be uncorrelated. A dataset of 2×10^5 samples are mixed with SNR values of 0dB, 20dB. The model is trained on a Linux GPU server with PCIe interface.

The DoA estimation accuracy is evaluated using the root mean squared error (RMSE) in degrees, which is defined as

$$\text{RMSE} = \sqrt{\mathbb{E} \left[\frac{1}{d} \sum_{i=1}^d (\theta_i - \hat{\theta}_i)^2 \right]}.$$

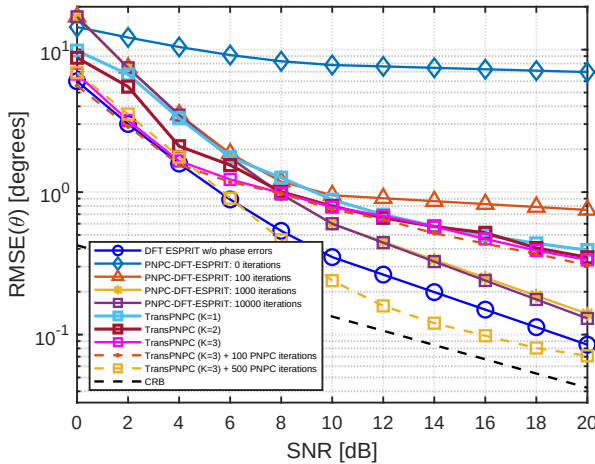


Fig. 3: RMSE vs. SNR

For performance comparison, we consider two baseline methods: The DFT-ESPRIT algorithm without phase errors and the iterative PNPC-DFT-ESPRIT algorithm with the impact of phase errors. We use different numbers of PNPC iterations to study the convergence behavior.

Fig. 3 illustrates the RMSE performance of different algorithms as a function of the SNR. In addition, the deterministic CRB [11] without phase errors is included. In the presence of phase errors, the PNPC-DFT-ESPRIT algorithm significantly improves the estimation performance as the number of iterations increases. However, achieving the performance of DFT-ESPRIT without the impact of phase errors requires a large

number of iterations, which leads to a high computational burden. The proposed TransPNPC ($K = 3$) method achieves a performance gain over the PNPC-DFT-ESPRIT with only 100 iterations. Regarding different number of unfolding blocks K , increasing K from $K = 1$ to $K = 3$ improves the estimation accuracy, especially in the SNR range from 0 to 10 dB. Furthermore, TransPNPC ($K = 3$) is used as an initial estimation stage, where its output is fed into the PNPC-DFT-ESPRIT algorithm for further model-based refinement. Specifically, TransPNPC ($K = 3$) combined with 100 PNPC iterations provides a slight performance improvement, while increasing the number of PNPC iterations to 500 provides a significant gain over the entire SNR range from 0 to 20 dB, even outperforming the DFT-ESPRIT baseline without phase errors. This behavior is due to the fact that TransPNPC is trained with noise-free and phase-error-free signal subspace labels, which enables the network to learn an implicit denoising for subspace estimation.

TABLE II: Computational Complexity Comparison.

Method	GFLOPs	Training Time
PNPC (100 iterations)	0.13 GFLOPs	—
TransPNPC (K=1)	0.016 GFLOPs	8.5 hours
TransPNPC (K=2)	0.033 GFLOPs	16.7 hours
TransPNPC (K=3)	0.049 GFLOPs	22.6 hours

Table II reports the computational complexity in terms of Giga Floating point operations (GFLOPs) [12] and the offline training time of the proposed TransPNPC method. Compared with the iterative PNPC algorithm with 100 iterations, TransPNPC reduces the inference complexity.

The results demonstrate that the proposed TransPNPC framework accelerates the convergence of the PNPC algorithm and effectively improves the subspace estimation quality through a learned denoising effect. Although TransPNPC requires an offline training cost, this is performed only once. After training, TransPNPC uses only a few unfolding blocks for new observations, while the conventional PNPC method may need hundreds of iterations. Therefore, for repeated real-time DoA estimation, TransPNPC has a lower online computational cost.

V. CONCLUSIONS

In this paper, we proposed TransPNPC, a transformer-based deep unfolding framework for phase-error compensation in DFT beamspace DoA estimation. By integrating data-driven phase-error estimation with model-based PNPC refinement, TransPNPC achieves fast convergence, and an improved estimation accuracy. Our simulation results demonstrate the effectiveness of the proposed approach in phase-error-distorted hybrid beamforming systems. The current TransPNPC architecture assumes a fixed number of sources d . As a future improvement, the beamspace covariance matrix can be used as the network input to support arbitrary source numbers. In addition, training using multiple SoIs may improve robustness under dynamic beam switching environment.

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